

RESfM from Scratch: Two Checkpoint-Selection Rules

Five training seeds (20–24), identical recipe, identical tracks — only the validation criterion that picks the

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The two rules

Both rows are the *same* supervised RESfM training (27 MegaDepth scenes, same schedule, same tracks, same robust BA). They differ only in which epoch is kept: **(A) ours** minimises **our_repro**, the label-free reprojection error of the network’s predicted points under its predicted cameras; **(B) authors’** maximises **Accuracy**, the outlier head’s classification agreement with the COLMAP-derived GT labels. MegaDepth is in-distribution; the other five are out-of-distribution.

Post-BA translation error

Dataset	translation mean±std		rotation / registered	
	(A) ours	(B) authors’	(A) rot / N_r	(B) rot / N_r
MegaDepth (in-dist)	0.365±0.125	0.318 ±0.106	1.89° / 85%	2.31° / 85%
Olsson	7.374 ±2.628	7.532±2.916	12.62° / 89%	12.94° / 89%
Strecha	0.388±0.724	0.383 ±0.662	1.56° / 99%	2.25° / 100%
BlendedMVS	0.353±0.043	0.348 ±0.032	31.66° / 92%	29.89° / 91%
1DSfM	10.498 ±1.447	11.120±1.538	9.62° / 77%	9.53° / 78%
1DSfM-hard	19.158±3.423	17.929 ±3.639	27.30° / 34%	28.32° / 39%

Per-seed values (seeds 20–24)

Dataset	rule	s20	s21	s22	s23	s24
MegaDepth (in-dist)	ours	0.344	0.189	0.423	0.529	0.340
	authors’	0.496	0.260	0.300	0.310	0.223
Olsson	ours	8.465	8.657	8.193	8.863	2.694
	authors’	8.765	8.657	2.322	8.891	9.025
Strecha	ours	0.006	0.222	0.036	1.674	0.005
	authors’	0.144	0.005	0.205	1.556	0.006
BlendedMVS	ours	0.379	0.280	0.348	0.387	0.370
	authors’	0.367	0.352	0.304	0.330	0.385
1DSfM	ours	11.032	9.954	8.521	12.468	10.518
	authors’	9.754	11.610	13.297	11.411	9.529
1DSfM-hard	ours	16.218	19.135	16.584	24.778	19.074
	authors’	17.575	15.386	16.579	24.267	15.836

Paired test (same seeds, so pairing is valid)

Dataset	mean diff (ours – authors’)	sd	$t(4)$	ours better on
MegaDepth (in-dist)	+0.047	0.153	+0.69	2/5 seeds
Olsson	-0.158	4.316	-0.08	3/5 seeds
Strecha	+0.005	0.164	+0.07	3/5 seeds
BlendedMVS	+0.005	0.052	+0.24	2/5 seeds
1DSfM	-0.622	2.615	-0.53	2/5 seeds
1DSfM-hard	+1.229	2.185	+1.26	1/5 seeds

$|t| > 2.78$ is required for $p < 0.05$ at four degrees of freedom; no dataset comes close.

What this shows

1. The rules are statistically indistinguishable everywhere. The authors’ criterion wins four datasets and ours wins two, every gap is a fraction of the corresponding seed-to-seed standard deviation, and the sign of the difference flips across datasets. The single-run pair quoted in the main paper (0.496 vs. 0.37 on MegaDepth) is seed 20, the most extreme of the five and the one most favourable to our rule; on the other four seeds the authors’ rule is better three times.

2. Both rules reproduce the Olsson collapse identically — four of five seeds at 8.2–9.0 with exactly one escape — but the escaping seed differs (s24 for ours, s22 for theirs). The collapse is a property of the supervised recipe; which seed dodges it is luck.

3. Both reproduce the one-bad-Strecha-seed pattern (s23: 1.674 vs. 1.556, with the other four seeds at 0.005–0.222), and both show the BlendedMVS rotation pathology (31.7° vs. 31.0°).

Conclusion. Deviating from the authors’ checkpoint rule does not advantage us: if anything their rule gives their own baseline a marginally better band, so every margin we report against the supervised baseline is a lower bound. Equally important, the failure modes survive a complete change of selection criterion, so they belong to the supervised recipe rather than to how we picked the epoch.

Cross-generalisation: train on each domain, evaluate on all

Each row is a training pool, each column an evaluation dataset; cells are post-BA translation, mean $\pm$ std over five training seeds. Diagonal cells are omitted (the pool’s own held-out split is the in-distribution number reported elsewhere). The two grids use the *same* pools and seeds and differ only in the outlier mechanism: supervised classification against GT-derived labels, versus label-free MADWEIGHT.

Supervised RESfM (GT-derived labels)

train pool	MD	Olsson	Strecha	BMVS	1DSfM	hard
MD-27	0.37 $\pm$ 0.12	7.37 $\pm$ 2.63	0.39 $\pm$ 0.72	0.35 $\pm$ 0.04	10.50 $\pm$ 1.45	19.16 $\pm$ 3.42
olssonid	0.62 $\pm$ 0.12	—	0.72 $\pm$ 0.98	6.70 $\pm$ 4.58	12.04 $\pm$ 3.19	20.17 $\pm$ 4.38
bmvsid	0.51 $\pm$ 0.10	6.95 $\pm$ 4.25	1.44 $\pm$ 0.32	—	10.45 $\pm$ 2.49	18.59 $\pm$ 0.48
strechaid	0.54 $\pm$ 0.03	8.62 $\pm$ 0.24	—	8.08 $\pm$ 1.23	10.22 $\pm$ 1.70	18.97 $\pm$ 0.22
1donly	0.56 $\pm$ 0.20	7.20 $\pm$ 3.45	1.92 $\pm$ 0.48	5.81 $\pm$ 2.29	—	20.12 $\pm$ 1.15
hardonly	0.83 $\pm$ 0.22	7.78 $\pm$ 2.82	1.94 $\pm$ 0.50	5.68 $\pm$ 4.47	8.49 $\pm$ 1.62	—
1dsfmid	0.59 $\pm$ 0.23	4.96 $\pm$ 1.86	2.37 $\pm$ 0.89	6.69 $\pm$ 2.46	—	—

Self-supervised madweight (label-free)

train pool	MD	Olsson	Strecha	BMVS	1DSfM	hard
MD-27	0.51 $\pm$ 0.08	3.01 $\pm$ 0.34	2.65 $\pm$ 0.25	0.14 $\pm$ 0.02	11.04 $\pm$ 1.68	18.92 $\pm$ 4.05
olssonid	0.52 $\pm$ 0.09	—	3.05 $\pm$ 0.34	6.97 $\pm$ 2.56	11.84 $\pm$ 1.91	15.56 $\pm$ 0.41
bmvsid	0.63 $\pm$ 0.10	2.73 $\pm$ 0.29	2.56 $\pm$ 0.46	—	11.50 $\pm$ 2.17	19.70 $\pm$ 1.84
strechaid	0.53 $\pm$ 0.19	2.69 $\pm$ 0.01	—	7.29 $\pm$ 2.68	13.94 $\pm$ 2.91	17.06 $\pm$ 1.39
1donly	0.53 $\pm$ 0.11	3.27 $\pm$ 0.21	2.94 $\pm$ 0.08	6.48 $\pm$ 2.29	—	18.31 $\pm$ 1.94
hardonly	0.47 $\pm$ 0.05	2.93 $\pm$ 0.49	3.02 $\pm$ 0.47	3.65 $\pm$ 3.02	13.21 $\pm$ 1.43	—
1dsfmid	0.52 $\pm$ 0.06	3.34 $\pm$ 0.41	2.98 $\pm$ 0.19	7.26 $\pm$ 1.26	—	—

What the grids show

- 1. The supervised Olsson collapse is training-domain-invariant.** Supervision degrades on Olsson from *every* training pool (4.96–8.62), while MADWEIGHT lands in a tight 2.69–3.34 band from every pool. The collapse is a property of the recipe, not an artefact of training on MegaDepth.
- 2. The Strecha exception is equally invariant, in the opposite direction.** Supervision wins Strecha from every pool (0.39–2.37 vs. 2.56–3.05) — the one dataset where labels help wherever they come from. It is also the smallest benchmark (54 cameras).
- 3. At the 60% target, label-free wins from anywhere.** The best supervised cell is 18.59; MADWEIGHT ranges 15.56–19.70 across pools, and reaches 12.1 when trained in-domain. No training pool rescues supervision at the extreme.
- 4. MegaDepth as a test set is nearly pool-independent** for both recipes (0.37–0.83 supervised, 0.47–0.63 label-free), suggesting its curated test scenes are easy for any reasonably trained model of this family.
- 5. One counterpoint worth stating.** Supervision trained on 1DSfM-hard transfers to 1DSfM at 8.49 — its best cross-domain cell anywhere, and better than MADWEIGHT from the same pool (13.21). Where labels are sound and the domains are related, supervision still transfers well.